

WANDRILLE RUFFENACH

EDUCATION

ENS Lyon

Since october 2024

Phd at LPENSL advised by Laurent Chevillard & Nikolay Tzvetkov.

ENS Lyon

September 2020 - July 2023

Physics master and bachelor. Bachelor and master obtained with highest honour.

SKILLS

Languages: French, English (fluent), German, Spanish, Italian (basics)

Programming: Python, Matlab, Julia

RESEARCH EXPERIENCE

Internship, INPHYNI Nice:

March 2024 - August 2024

Anomalous diffusion by fractal homogenization: A numerical exploration. *Advised by* [Éric Simonnet](#)

Internship, SPEC CEA Saclay:

September 2023 - February 2024

Cheap turbulence modeling with 3d point vortices coupled to a shear flow. *Advised by* [Bérengère Dubrulle](#)

M2 internship, ENS Lyon:

January 2023 - July 2023

Numerical study of a linear model of turbulent cascade. *Advised by* [Laurent Chevillard](#)

M1 internship, Newcastle University:

May 2022 - July 2022

Superfluid drain vortex. *Advised by* [Prof Carlo .F Barenghi](#)

Bachelor internship, ENS Lyon:

june 2021 - July 2021

LIF measurements of temperature field in Rayleigh-Bénard convection. *Advised by* [Julien Salort](#)

TEACHING EXPERIENCE

Introduction to signal & image processing.

2024-2026

Bachelor students, ENS de Lyon

Training for the "Agrégation", french qualifying exam for teaching.

2024-2026

Bachelor students, ENS de Lyon

Tutoring in CPES

2024-2026

CPES students, ENS de Lyon

Tutoring in Quantum mechanics & mathematics

2024-2026

Bachelor students, ENS de Lyon

Oral examination for first and second year physics students

2021 - 2023

Preparatory Classes (PCSI, PC & PSI) Les Chartreux, Lyon

PUBLICATIONS

A causal random synthetic turbulent velocity field: The underlying Gaussian structure Matthieu Chatelain, Júlia Domingues Lemos, Wandrille Ruffenach, Mickaël Bourgoïn, Charles-Edouard Bréhier, Laurent Chevillard, Ilias Sibgatullin, Romain Volk *arxiv pre-print* 2025

Spontaneous stochasticity in the Armstrong-Vicol passive scalar W. Ruffenach, E. Simonnet and N. Valade *arxiv pre-print* 09/2025

Spontaneous stochasticity and the Armstrong-Vicol passive scalar W. Ruffenach, E. Simonnet and N. Valade *arxiv pre-print* 04/2025

A linear model of dynamical sea wavenumber spectra G. Beck, C-E. Bréhier, L. Chevillard, R. Grande and W. Ruffenach. *arxiv pre-print* 2025

Subgrid-scale modeling of turbulent shear flow with vortons. W. Ruffenach, L. Fery and B. Dubrulle. *Phys. Rev. Fluids*. 2025

Numerical simulations of a stochastic dynamics leading to cascades and loss of regularity: Applications to fluid turbulence and generation of fractional Gaussian fields. G. Beck, C-E. Bréhier, L. Chevillard, R. Grande and W. Ruffenach. *Phys. Rev. Res.* 6(3) 2024

Superfluid drain vortex. W. Ruffenach, L. Galantucci and C. F. Barenghi. *J. Low Temp. Phys.* 2023

TALKS

A linear and stochastic model of energy cascade. Theory Day, LPENSL, 05/2025.

A linear and stochastic model of energy cascade. GdR Turblence NS 2.0, Marseille, 06/2025.

POSTERS

Anomalous diffusion and spontaneous stochasticity in collapsing wave turbulence. Fall school Bridges over turbulent matters, IESC Cargèse, 04/2025.

A linear and stochastic model of energy cascade. Summer school Mathematical fluid dynamics, IESC Cargèse, 10/2025.

Cheap turbulence modeling with vortex particles. Rencontres du Non-linéaire, 03/2024.